



BobSim

Ramp-Steer Vehicle Characterization

Measured- a_y ramp-response isolines with robust metric fits

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Run Notes

- Open-loop constant-rate handwheel ramp at the configured test velocities
- steerStart = 2.0 s; trim all pre-ramp samples
- Measured a_y is the analysis axis
- Metrics describe the ramp-response path, not fully settled equilibrium holds
- $|a_y| < 1.0 \text{ m/s}^2$ is excluded; linear band = 1.0 to 4.0 m/s^2
- Ramp terminates when local lateral gain drops 20% from the linear reference
- Limit metrics are local derivatives at the largest measured $|a_y|$ reached
- Tire-lift protection terminates if any tire falls below the configured normal-load threshold
- Raw plots are rendered; summary metrics use robust smoothed derivatives

SteadyStateEval Summary 12.5 m/s

Metric	Value	Units
Measured a_y Range	1.00 → 18.74	m/s^2
Linear Ramp Sideslip Gradient	0.00178	$\frac{\text{rad}}{\text{m/s}^2}$
	1.00	$\frac{\text{deg}}{g}$
Limit Ramp Sideslip Gradient	-0.00510	$\frac{\text{rad}}{\text{m/s}^2}$
	-2.87	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00002	$\frac{\text{rad}}{\text{m/s}^2}$
	0.01	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	-0.00191	$\frac{\text{rad}}{\text{m/s}^2}$
	-1.08	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.04192	$\frac{\text{rad}}{\text{m/s}^2}$
	23.56	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.04580	$\frac{\text{rad}}{\text{m/s}^2}$
	25.74	$\frac{\text{deg}}{g}$
Linear Ramp Roll Gradient	0.890	$\frac{\text{deg}}{g}$
Limit Ramp Roll Gradient	0.812	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-21.6	$\text{N} \cdot \text{m}$
Peak Torque a_y	18.09	m/s^2

SteadyStateEval Summary 15.0 m/s

Metric	Value	Units
Measured a_y Range	1.00 → 19.10	m/s^2
Linear Ramp Sideslip Gradient	0.00044	$\frac{\text{rad}}{\text{m/s}^2}$
	0.25	$\frac{\text{deg}}{g}$
Limit Ramp Sideslip Gradient	−0.00620	$\frac{\text{rad}}{\text{m/s}^2}$
	−3.48	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00005	$\frac{\text{rad}}{\text{m/s}^2}$
	0.03	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	−0.00106	$\frac{\text{rad}}{\text{m/s}^2}$
	−0.60	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.02940	$\frac{\text{rad}}{\text{m/s}^2}$
	16.52	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.03282	$\frac{\text{rad}}{\text{m/s}^2}$
	18.45	$\frac{\text{deg}}{g}$
Linear Ramp Roll Gradient	0.892	$\frac{\text{deg}}{g}$
Limit Ramp Roll Gradient	0.834	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-21.5	$\text{N} \cdot \text{m}$
Peak Torque a_y	18.52	m/s^2

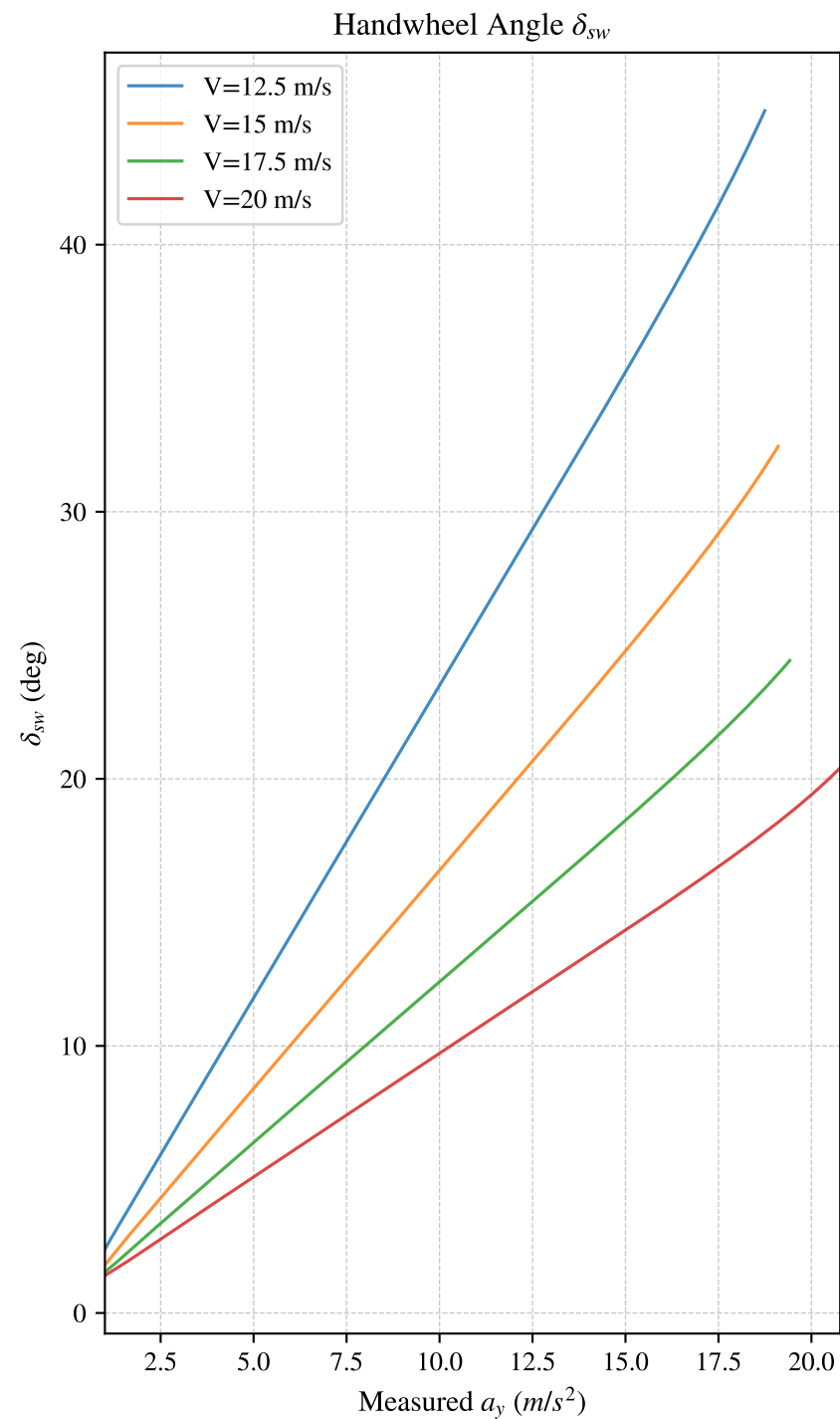
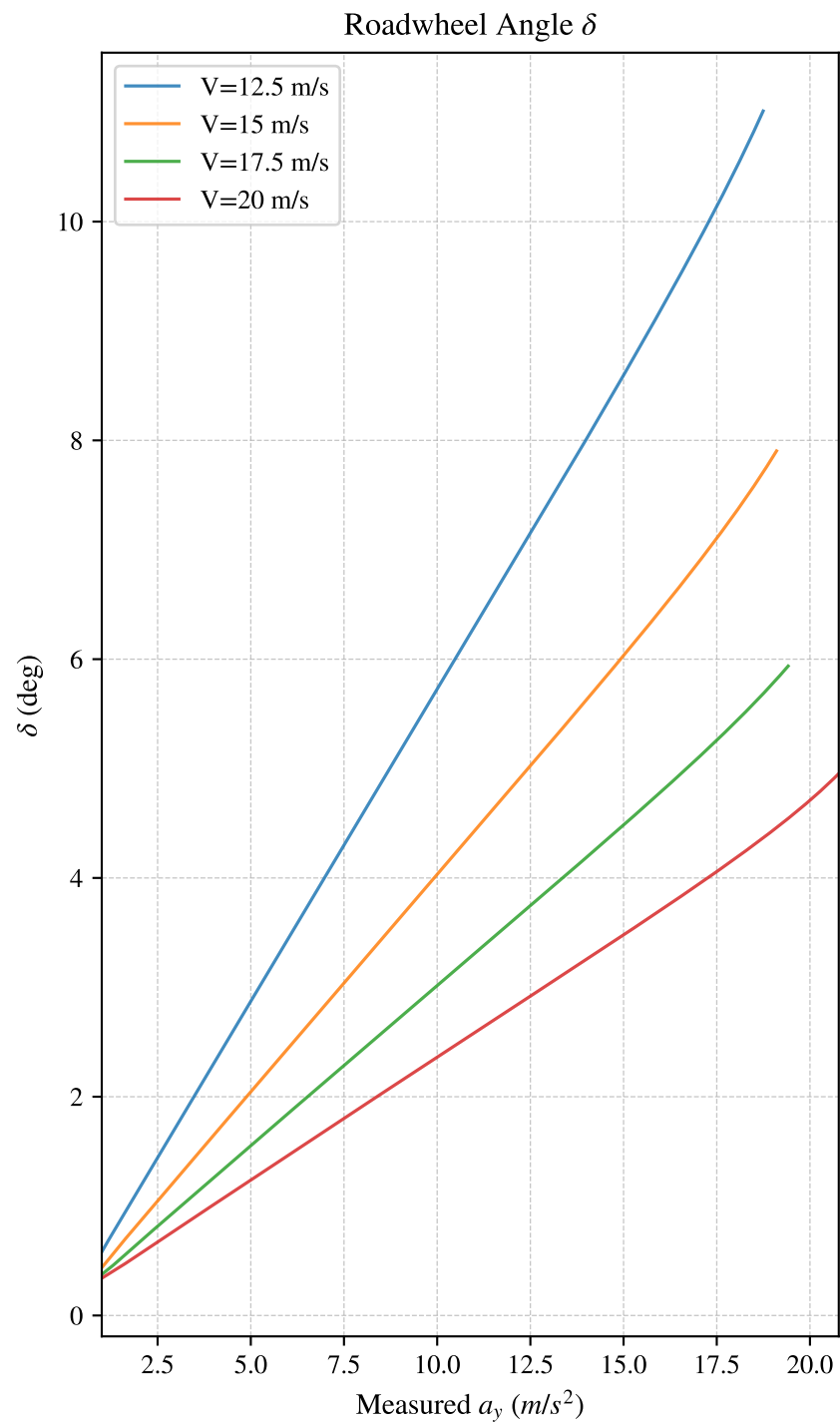
SteadyStateEval Summary 17.5 m/s

Metric	Value	Units
Measured a_y Range	1.00 → 19.42	m/s^2
Linear Ramp Sideslip Gradient	-0.00030	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.17	$\frac{\text{deg}}{g}$
Limit Ramp Sideslip Gradient	-0.00688	$\frac{\text{rad}}{\text{m/s}^2}$
	-3.87	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00002	$\frac{\text{rad}}{\text{m/s}^2}$
	0.01	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	-0.00081	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.46	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.02163	$\frac{\text{rad}}{\text{m/s}^2}$
	12.16	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.02410	$\frac{\text{rad}}{\text{m/s}^2}$
	13.55	$\frac{\text{deg}}{g}$
Linear Ramp Roll Gradient	0.898	$\frac{\text{deg}}{g}$
Limit Ramp Roll Gradient	0.861	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-21.8	$\text{N} \cdot \text{m}$
Peak Torque a_y	19.01	m/s^2

SteadyStateEval Summary 20.0 m/s

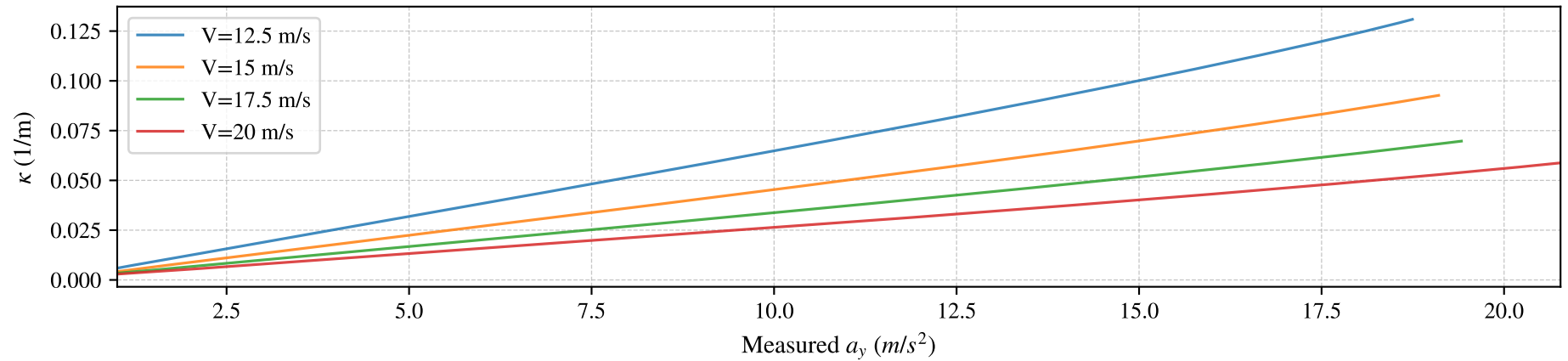
Metric	Value	Units
Measured a_y Range	1.01 → 20.77	m/s^2
Linear Ramp Sideslip Gradient	-0.00097	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.55	$\frac{\text{deg}}{g}$
Limit Ramp Sideslip Gradient	-0.00853	$\frac{\text{rad}}{\text{m/s}^2}$
	-4.79	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00003	$\frac{\text{rad}}{\text{m/s}^2}$
	0.02	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	-0.00080	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.45	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.01679	$\frac{\text{rad}}{\text{m/s}^2}$
	9.44	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.01899	$\frac{\text{rad}}{\text{m/s}^2}$
	10.67	$\frac{\text{deg}}{g}$
Linear Ramp Roll Gradient	0.905	$\frac{\text{deg}}{g}$
Limit Ramp Roll Gradient	0.887	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-22.5	$\text{N} \cdot \text{m}$
Peak Torque a_y	19.58	m/s^2

Raw Steering Samples vs Measured a_y

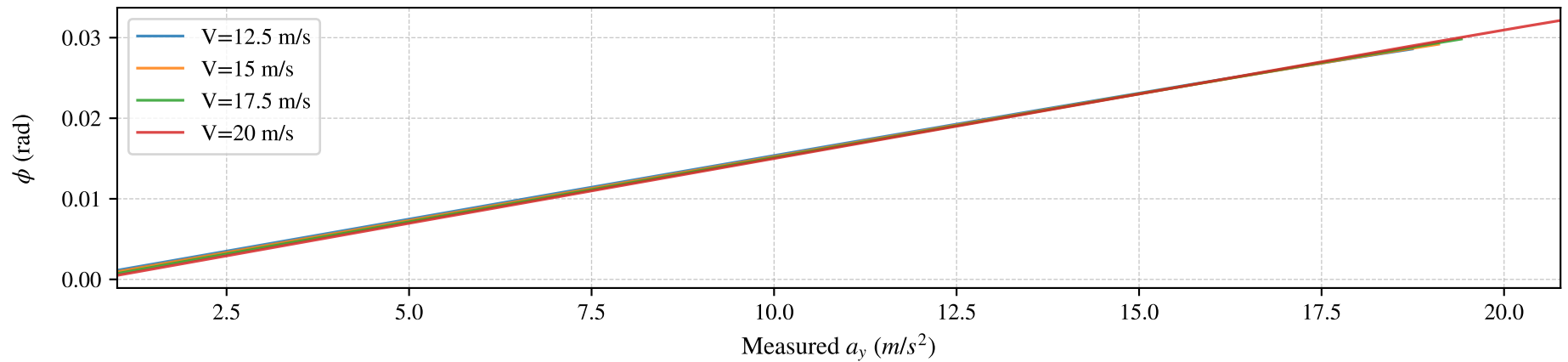


Raw Vehicle State Samples vs Measured a_y

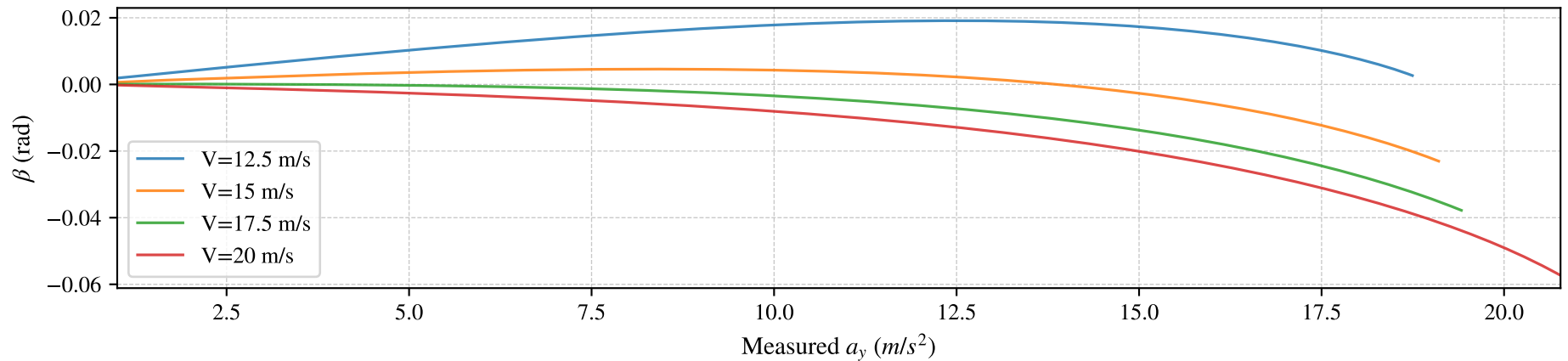
Curvature κ



Roll Angle ϕ



Sideslip β



Raw Steering Effort Samples vs Measured a_y

